



100V N-Channel NexFET™ Power MOSFETs

Check for Samples: [CSD19531KCS](#)

FEATURES

- Ultra Low Qg and Qgd
- Low Thermal Resistance
- Avalanche Rated
- Pb Free Terminal Plating
- RoHS Compliant
- Halogen Free
- TO-220 Plastic Package

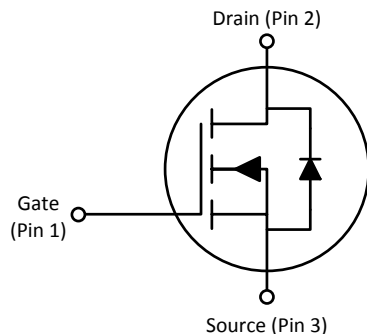
APPLICATIONS

- Secondary Side Synchronous Rectifier
- Hot Swap Telecom
- Motor Control

DESCRIPTION

This 100V, 6.4mΩ, TO-220 NexFET™ power MOSFET has been designed to minimize losses in power conversion applications.

Pin Out Drawing



PRODUCT SUMMARY

T _A = 25°C		TYPICAL VALUE	UNIT
V _{DS}	Drain to Source Voltage	100	V
Q _g	Gate Charge Total (10V)	37	nC
Q _{gd}	Gate Charge Gate to Drain	7.5	nC
R _{DS(on)}	Drain to Source On Resistance	V _{GS} = 6V	7.3 mΩ
		V _{GS} = 10V	6.4 mΩ
V _{GS(th)}	Threshold Voltage	2.7	V

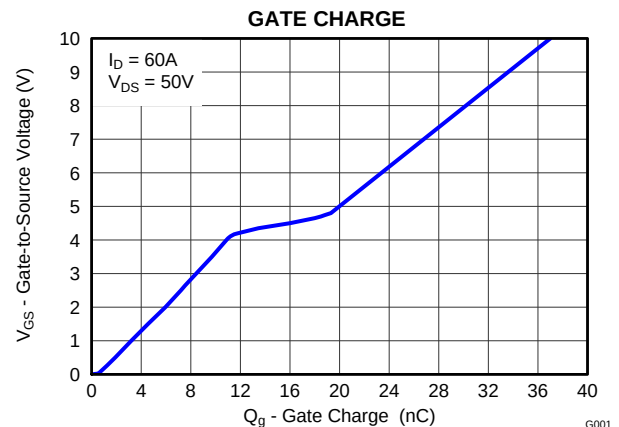
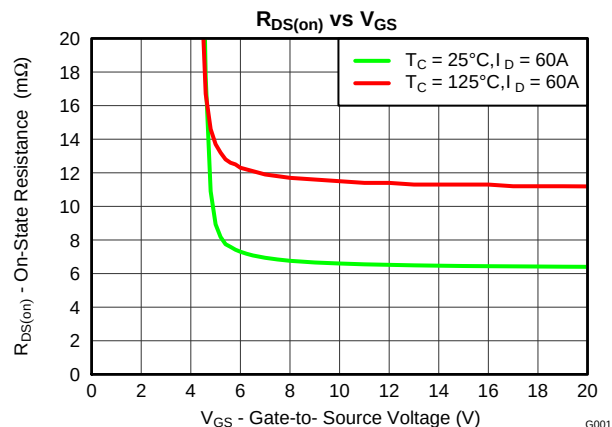
ORDERING INFORMATION

Device	Package	Media	Qty	Ship
CSD19531KCS	TO-220 Plastic Package	Tube	50	Tube

ABSOLUTE MAXIMUM RATINGS

T _A = 25°C		VALUE	UNIT
V _{DS}	Drain to Source Voltage	100	V
V _{GS}	Gate to Source Voltage	±20	V
I _D	Continuous Drain Current (Package limited), T _C = 25°C	100	A
	Continuous Drain Current (Silicon limited), T _C = 25°C	105	
	Continuous Drain Current (Silicon limited), T _C = 100°C	67	
I _{DM}	Pulsed Drain Current ⁽¹⁾	122	A
P _D	Power Dissipation	179	W
T _J , T _{STG}	Operating Junction and Storage Temperature Range	–55 to 150	°C
E _{AS}	Avalanche Energy, single pulse I _D = 60A, L = 0.1mH, R _G = 25Ω	180	mJ

(1) Pulse duration ≤300μs, Duty cycle ≤1%



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These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

ELECTRICAL CHARACTERISTICS

($T_A = 25^\circ\text{C}$ unless otherwise stated)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
Static Characteristics						
BV _{DSS}	Drain to Source Voltage	V _{GS} = 0V, I _D = 250μA	100			V
I _{DSS}	Drain to Source Leakage Current	V _{GS} = 0V, V _{DS} = 80V			1	μA
I _{GSS}	Gate to Source Leakage Current	V _{DS} = 0V, V _{GS} = 20V			100	nA
V _{GS(th)}	Gate to Source Threshold Voltage	V _{DS} = V _{GS} , I _D = 250μA	2.2	2.7	3.3	V
R _{DS(on)}	Drain to Source On Resistance	V _{GS} = 6V, I _D = 60A		7.3	8.8	mΩ
		V _{GS} = 10V, I _D = 60A		6.4	7.7	mΩ
g _{fs}	Transconductance	V _{DS} = 10V, I _D = 60A		137		S
Dynamic Characteristics						
C _{iss}	Input Capacitance	V _{GS} = 0V, V _{DS} = 50V, f = 1MHz		2980	3870	pF
C _{oss}	Output Capacitance			560	728	pF
C _{rss}	Reverse Transfer Capacitance			13	17	pF
R _G	Series Gate Resistance			1.3	2.6	Ω
Q _g	Gate Charge Total (10V)	V _{DS} = 50V, I _D = 60A		38		nC
Q _{gd}	Gate Charge Gate to Drain			7.5		nC
Q _{gs}	Gate Charge Gate to Source			11.9		nC
Q _{g(th)}	Gate Charge at V _{th}			7.3		nC
Q _{oss}	Output Charge	V _{DS} = 50V, V _{GS} = 0V		98		nC
t _{d(on)}	Turn On Delay Time	V _{DS} = 50V, V _{GS} = 10V, I _{DS} = 60A, R _G = 0Ω		8.4		ns
t _r	Rise Time			7.2		ns
t _{d(off)}	Turn Off Delay Time			16		ns
t _f	Fall Time			4.1		ns
Diode Characteristics						
V _{SD}	Diode Forward Voltage	I _{SD} = 60A, V _{GS} = 0V		0.9	1	V
Q _{rr}	Reverse Recovery Charge	V _{DS} = 50V, I _F = 60A, di/dt = 300A/μs		270		nC
t _{rr}	Reverse Recovery Time			83		ns

THERMAL CHARACTERISTICS

($T_A = 25^\circ\text{C}$ unless otherwise stated)

PARAMETER		MIN	TYP	MAX	UNIT
$R_{\theta JC}$	Thermal Resistance Junction to Case			0.7	$^\circ\text{C/W}$
$R_{\theta JA}$	Thermal Resistance Junction to Ambient			62	$^\circ\text{C/W}$

TYPICAL MOSFET CHARACTERISTICS

($T_A = 25^\circ\text{C}$ unless otherwise stated)

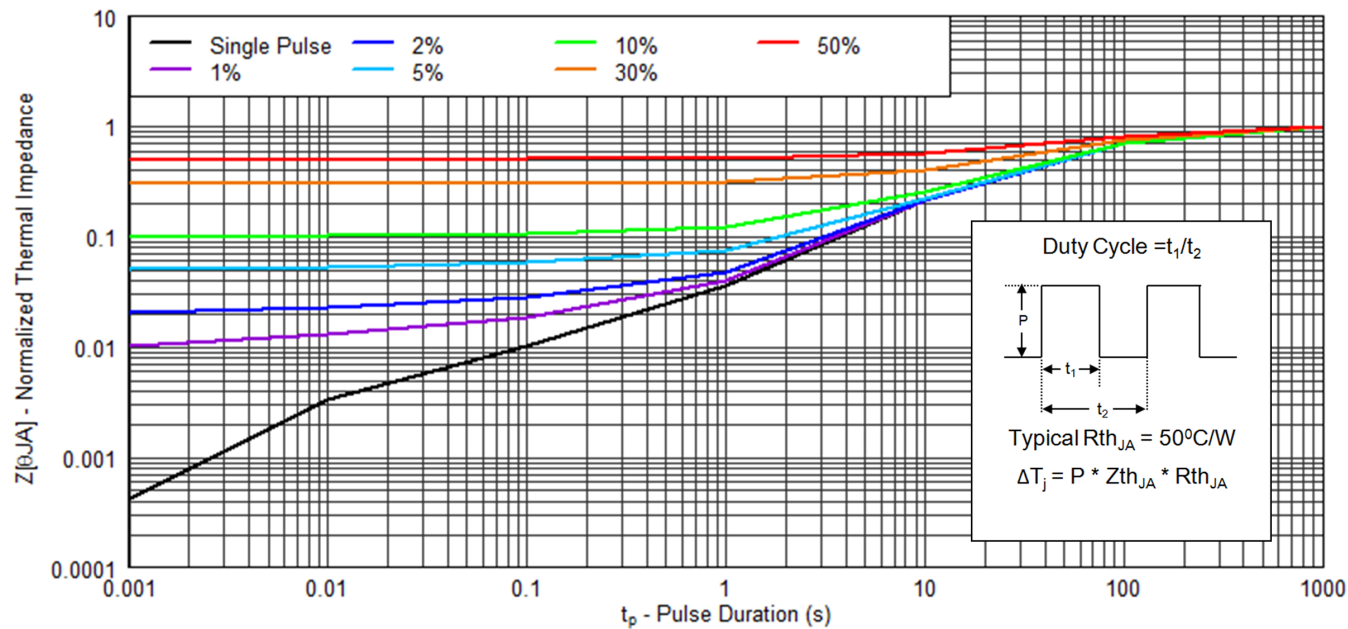


Figure 1. Transient Thermal Impedance

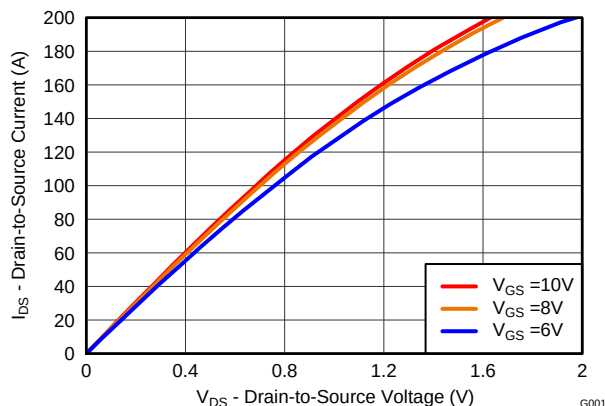


Figure 2. Saturation Characteristics

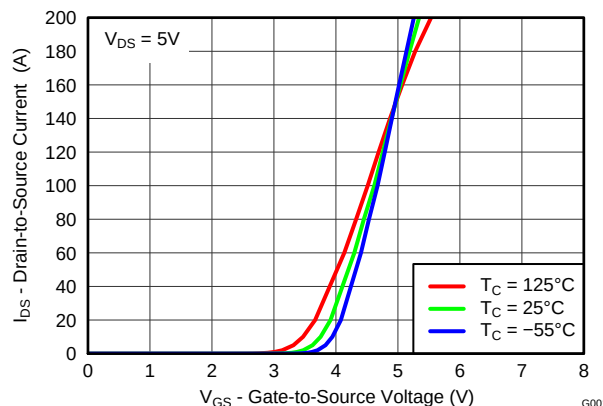


Figure 3. Transfer Characteristics

TYPICAL MOSFET CHARACTERISTICS (continued)

($T_A = 25^\circ\text{C}$ unless otherwise stated)

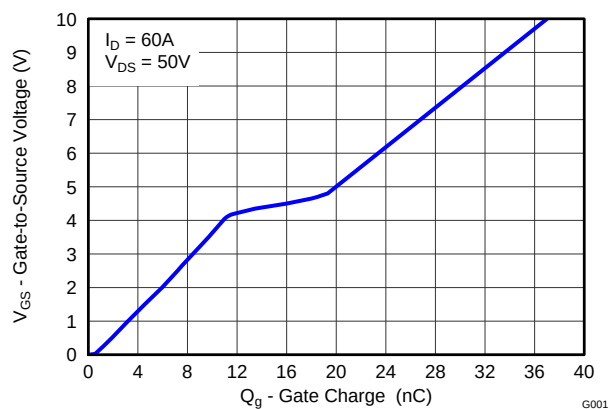


Figure 4. Gate Charge

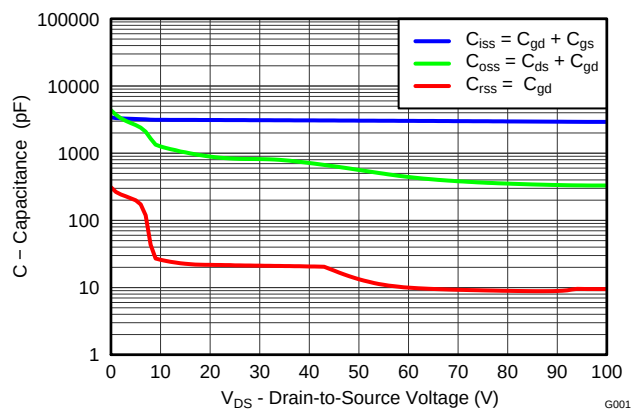


Figure 5. Capacitance

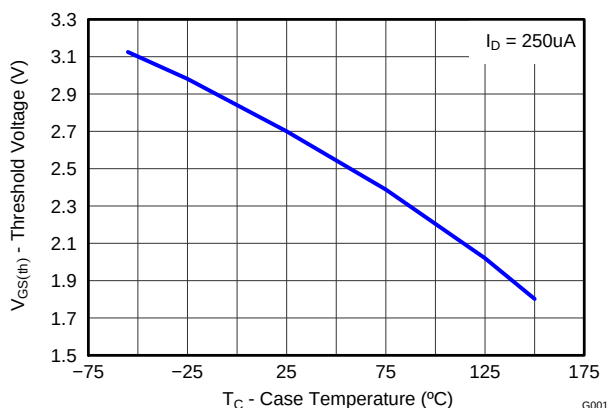


Figure 6. Threshold Voltage vs. Temperature

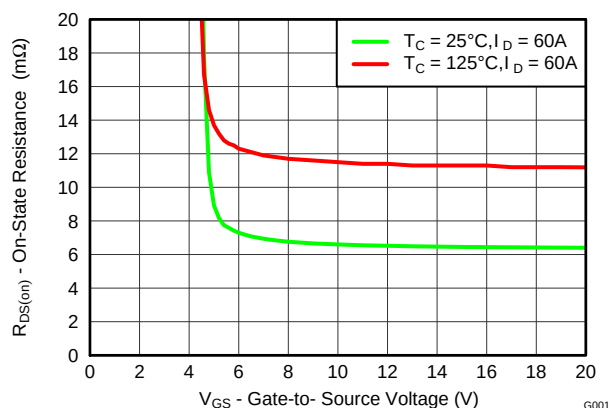


Figure 7. On-State Resistance vs. Gate-to-Source Voltage

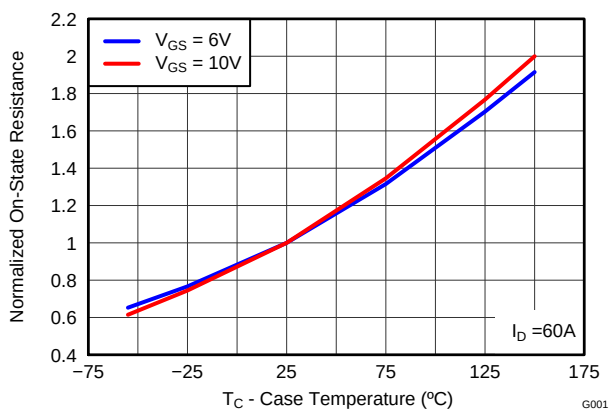


Figure 8. Normalized On-State Resistance vs. Temperature

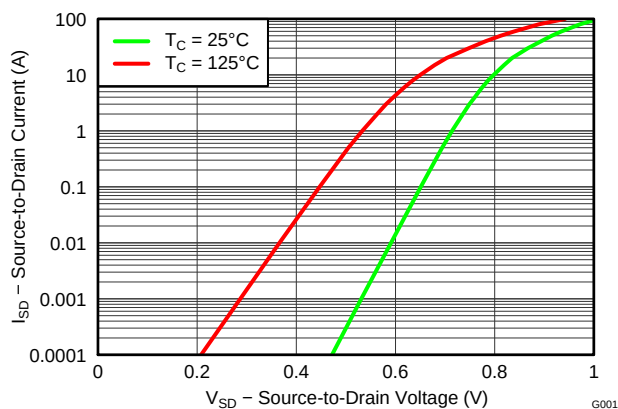


Figure 9. Typical Diode Forward Voltage

TYPICAL MOSFET CHARACTERISTICS (continued)

($T_A = 25^\circ\text{C}$ unless otherwise stated)

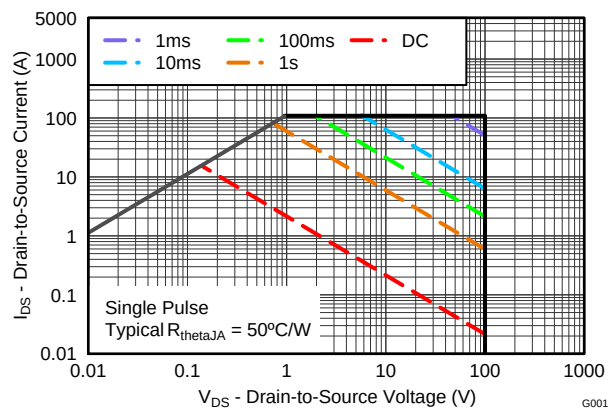


Figure 10. Maximum Safe Operating Area

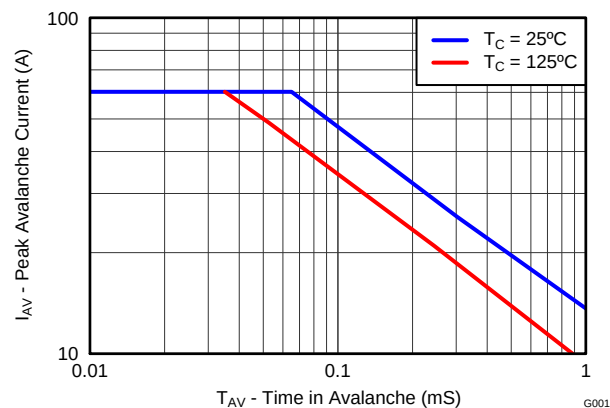


Figure 11. Single Pulse Unclamped Inductive Switching

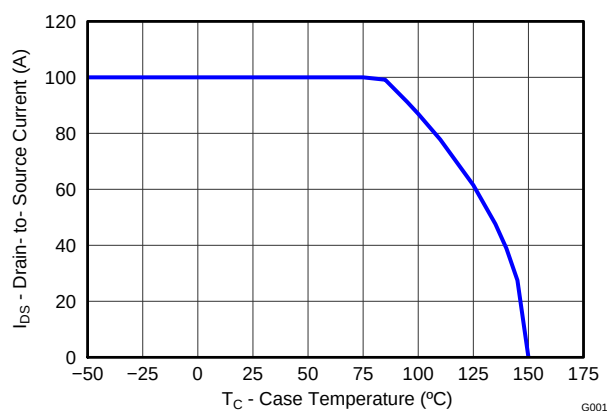
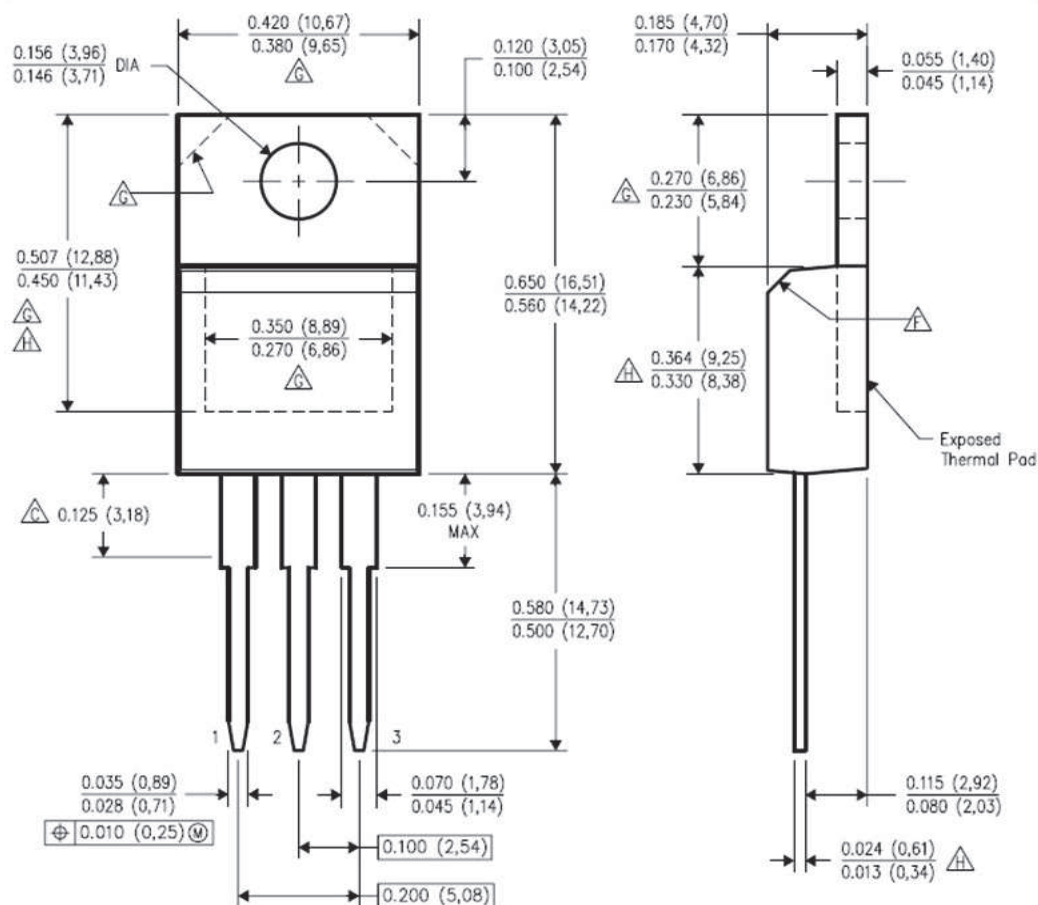


Figure 12. Maximum Drain Current vs. Temperature

MECHANICAL DATA

KCS Package Dimensions



Notes:

1. All linear dimensions are in inches
2. This drawing is subject to change without notice
3. Lead Dimensions are not controlled within "C" area
4. All lead dimensions apply before solder dip
5. The center lead is in electrical contact with the mounting tab
6. The chamfer at "F" is optional
7. Thermal pad contour at "G" optional with these dimensions
8. "H" Falls within JEDEC TO-220 variation AB, except minimum lead thickness, minimum exposed pad length, and maximum body length.

Table 1. Pin Configuration

Position	Designation
Pin 1	Gate
Pin 2 / Tab	Drain
Pin 3	Source

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